

Implementation and Performance Evaluation of Contrast Limited Adaptive Local Histogram Equalization for Poor Contrast Image Enhancement

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Abstract—Image contrast enhancement is an essential preprocessing step in digital image processing because low-quality images often contain weak visibility, poor illumination distribution, and loss of local detail. These problems reduce the usefulness of images in applications such as surveillance, medical imaging, and visual inspection. Conventional histogram equalization methods improve global contrast but frequently distort brightness and suppress fine image structures. Contrast Limited Adaptive Histogram Equalization (CLAHE) addresses some of these issues by performing local enhancement and limiting histogram amplification, yet its effectiveness depends strongly on manually selected parameters such as clip limit and contextual window size. In this work, an implementation of Contrast Limited Adaptive Local Histogram Equalization (CLALHE) is presented in MATLAB for the enhancement of poor-contrast grayscale images. The method adaptively determines enhancement parameters from histogram peaks and valleys, evaluates candidate settings through a quality-driven fitness function, and applies local enhancement on subdivided image regions. The proposed implementation was tested on three sample low-contrast images and compared with conventional CLAHE. Experimental observations show that the implemented CLALHE approach provides better local detail visibility, improved structural preservation, and reduced brightness distortion in most cases. Quantitative evaluation using PSNR, SSIM, entropy, and AMBE supports the effectiveness of the method.

Index Terms—image enhancement, histogram equalization, CLAHE, CLALHE, contrast enhancement, MATLAB

I. INTRODUCTION

Digital images acquired under nonideal conditions often suffer from poor visibility due to low illumination, limited dynamic range, sensor limitations, and environmental effects. Such degradations reduce the interpretability of important features and make subsequent processing stages, including segmentation, recognition, and detection, less reliable. Therefore, contrast enhancement remains one of the most important preprocessing tasks in digital image processing.

Histogram-based methods are popular because of their simplicity and low computational cost. Classical histogram equalization redistributes intensity levels across the full available range, thereby improving global contrast. However, this method may significantly alter image brightness and introduce an unnatural appearance. Adaptive Histogram Equalization (AHE) improves local contrast by operating on small neighborhoods, but it tends to amplify noise in homogeneous regions. CLAHE was introduced to control this problem by clipping

the local histogram before equalization. Although CLAHE performs better than many traditional techniques, it still suffers from important limitations. Its clip limit and tile size are typically selected manually, and a fixed enhancement setting is applied uniformly over the image even though different regions may contain different visual characteristics.

To address these limitations, recent research has focused on adaptive versions of CLAHE. One such method is Contrast Limited Adaptive Local Histogram Equalization (CLALHE), in which enhancement parameters are estimated automatically from image histogram characteristics, and enhancement is applied locally to subimages rather than uniformly over the full image.

In this paper, we present a MATLAB implementation of CLALHE and evaluate its performance on three input images with poor contrast. The main objective of this work is not to propose a new algorithm, but to implement the selected 2025 reference method in a practical and reproducible form, compare it with conventional CLAHE, and analyze its behavior using both visual and quantitative measures.

The major contributions of this work are as follows:

- 1) Implementation of the CLALHE enhancement framework in MATLAB.
- 2) Integration of adaptive parameter selection based on histogram peaks and valleys.
- 3) Performance comparison with conventional CLAHE using standard image quality metrics.
- 4) Visual analysis using three representative low-contrast input images.

The remainder of the paper is organized as follows. Section II reviews related work. Section III explains the implemented methodology. Section IV describes the experimental setup and evaluation metrics. Section V presents the results and discussion. Section VI concludes the paper.

II. RELATED WORK

Contrast enhancement has been studied extensively in both global and local forms. Global Histogram Equalization (GHE) is one of the earliest and simplest approaches. It stretches intensity values according to the cumulative histogram distribution and improves overall image contrast. Despite its simplicity, GHE often shifts average brightness and suppresses subtle local details, especially in images containing both dark and bright regions.

AHE was developed to overcome the global limitation of GHE by computing equalization over local neighborhoods. While it enhances local structures more effectively, it can also greatly amplify background noise, especially in nearly uniform areas. CLAHE modifies AHE by clipping the local histogram prior to equalization, thereby reducing excessive enhancement. Due to its balance between simplicity and effectiveness, CLAHE has become widely used in medical imaging, document analysis, surveillance, and consumer imaging systems.

Several researchers have proposed improved versions of CLAHE to reduce parameter sensitivity and artifact generation. Some methods introduce adaptive clip limit calculation, while others optimize contextual block size or combine CLAHE with gamma correction, homomorphic filtering, or entropy-based selection criteria. More recent methods such as IAECHE and AEIHE attempt to estimate enhancement settings iteratively using image information measures. However, these methods may require higher computation time or may not perform consistently across different image types.

The CLALHE method selected in this work extends the CLAHE framework in two main ways. First, it determines enhancement parameters adaptively using histogram peaks and valleys. Second, it divides the input image into subimages and enhances each part independently. This local-adaptive strategy is intended to preserve structural information while improving low-contrast regions more effectively than standard CLAHE.

III. METHODOLOGY

A. Overview

The implemented CLALHE algorithm consists of two main stages:

- 1) Adaptive parameter determination
- 2) Subimage-based local enhancement

The overall processing pipeline is as follows:

- read image,
- convert to grayscale if required,
- compute histogram,
- detect peaks and valleys,
- estimate clip limit and contextual block dimension,
- evaluate candidate parameter sets using a fitness measure,
- divide the image into subimages,
- apply adaptive CLAHE locally,
- merge the enhanced blocks to obtain the final image.

B. Histogram Analysis

Let the grayscale input image be represented by $I(x, y)$. Its histogram is computed over 256 intensity levels. A histogram peak is identified when the frequency of a gray level is greater than the frequencies of its neighboring gray levels. Similarly, a valley is identified when the frequency is smaller than its neighboring frequencies.

If the number of peaks is denoted by N_{peaks} and the number of valleys by $N_{valleys}$, these values are used to estimate the enhancement parameters.

C. Adaptive Parameter Estimation

An intermediate parameter is computed from the valley frequencies as

$$N_p = \frac{\sum Valley(i)}{N_{valleys}} \quad (1)$$

where $Valley(i)$ denotes the histogram count at the i -th valley.

The contextual block dimension (CBD) is then selected as

$$CBD = \lceil \ln(N_{peaks}) \rceil \quad (2)$$

where $\lceil \cdot \rceil$ represents the ceiling operator.

The ideal clip limit is estimated from peak and valley information as

$$I_{CL} = \frac{Peak(i) + N_p}{\max(Valley)} \quad (3)$$

where $Peak(i)$ is the frequency corresponding to a candidate histogram peak.

D. Parameter Selection Using CIQI

To determine the most suitable enhancement parameters, multiple candidate values are tested and assessed using a quality-based criterion. In our implementation, the following Comprehensive Image Quality Index (CIQI) is used:

$$CIQI = \frac{PSNR + Entropy}{AMBE} \quad (4)$$

A larger CIQI indicates better enhancement performance because it favors higher information content and lower brightness deviation. The candidate parameter set yielding the highest CIQI is selected for the second stage of enhancement.

E. Subimage-Based Enhancement

After optimal parameters are selected, the image is divided into a number of subimages determined from the number of valleys:

$$ONS \approx \lceil \log_2(N_{valleys}) \rceil \quad (5)$$

To maintain a balanced partition, the number of subimages is adjusted to an even number if necessary. Each subimage is then enhanced independently using MATLAB's CLAHE implementation with the selected clip limit and tile size. Finally, all enhanced subimages are concatenated to reconstruct the final output image.

F. MATLAB Implementation

The method was implemented using MATLAB with the Image Processing Toolbox. Three core modules were used:

- `main.m` for loading images, calling enhancement methods, and displaying results,
- `clalhe.m` for the adaptive CLALHE implementation,
- `compute_metrics.m` for PSNR, SSIM, entropy, and AMBE calculation.

For comparison, conventional CLAHE was implemented using `adapthisteq` with fixed baseline settings.

IV. EXPERIMENTAL SETUP

A. Test Images

Three low-contrast images were used for evaluation. These images represent different intensity distributions and local detail patterns. The input images are referred to as:

- Image 1: test.jpg
- Image 2: test1.jpg
- Image 3: test2.jpg

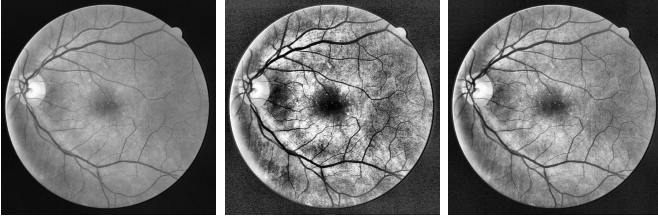


Fig. 1. Visual comparison for Image 1: original image, conventional CLAHE result, and CLALHE result.



Fig. 2. Visual comparison for Image 2: original image, conventional CLAHE result, and CLALHE result.



Fig. 3. Visual comparison for Image 3: original image, conventional CLAHE result, and CLALHE result.

B. Evaluation Metrics

The following quantitative measures were used.

1) *Peak Signal-to-Noise Ratio (PSNR)*: PSNR measures similarity between the original and enhanced image in decibels. A higher PSNR generally indicates less distortion.

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (6)$$

2) *Structural Similarity Index (SSIM)*: SSIM evaluates the preservation of luminance, contrast, and structure. Values closer to 1 indicate higher structural similarity.

3) *Entropy*: Entropy measures the richness of information in the enhanced image. Larger values usually indicate better detail representation, although high entropy alone does not always guarantee visually pleasing results.

4) *Absolute Mean Brightness Error (AMBE)*: AMBE measures the absolute difference between the mean intensity of the input and output images:

$$AMBE = |\mu_{in} - \mu_{out}| \quad (7)$$

A smaller AMBE indicates better brightness preservation.

V. RESULTS AND DISCUSSION

A. Qualitative Analysis

The visual comparison between CLAHE and CLALHE shows clear differences in enhancement behavior. In the CLAHE results, contrast is improved, but several regions appear overly sharpened or unnaturally bright. This effect is particularly noticeable in smooth background areas, where local amplification sometimes introduces an artificial appearance.

By contrast, the CLALHE results show a more balanced enhancement. The method improves local visibility while better preserving the natural brightness distribution of the image. Fine structures become more distinguishable, and the transition between enhanced regions appears smoother because the enhancement is adapted to histogram characteristics and applied locally over subdivided regions.

For Image 1, CLALHE produces a noticeable improvement in edge visibility and structural contrast while avoiding excessive brightness increase. For Image 2, the proposed implementation maintains better structural consistency and yields clearer local details than conventional CLAHE. For Image 3, CLALHE again provides stronger detail visibility and better visual balance, especially in darker areas where CLAHE tends to produce uneven enhancement.

B. Quantitative Analysis

Table I summarizes the numerical results obtained from MATLAB for the three test images.

TABLE I
QUANTITATIVE COMPARISON OF CLAHE AND CLALHE

Image	Method	PSNR	SSIM	Entropy	AMBE
Fig 1	CLAHE	16.1589	0.4536	7.8488	0.0393
	CLALHE	23.5792	0.8163	7.4746	0.0199
Fig 2	CLAHE	16.6475	0.6122	7.8349	0.0135
	CLALHE	19.5543	0.7180	7.8016	0.0303
Fig 3	CLAHE	17.8283	0.6860	7.1284	0.0956
	CLALHE	22.5501	0.8948	6.4415	0.0360

The results indicate that CLALHE consistently achieves higher PSNR and SSIM than conventional CLAHE for all three test images. This suggests that the enhanced images preserve structural information more effectively and introduce less distortion relative to the original.

For test.jpg, PSNR improves from 16.1589 dB to 23.5792 dB, while SSIM increases from 0.4536 to 0.8163. At the same time, AMBE decreases from 0.0393 to 0.0199, indicating improved brightness preservation. Although entropy is slightly lower for CLALHE, the visual quality is better, showing that controlled enhancement can be preferable to simply maximizing information spread.

For `test1.jpg`, CLALHE again improves PSNR and SSIM, but AMBE is slightly higher than for CLAHE. This suggests that while CLALHE preserves structure better, its adaptive local processing produces a somewhat larger mean brightness change in this particular image. Nevertheless, the improvement in visual clarity and structural similarity remains favorable.

For `test2.jpg`, CLALHE produces strong gains in both PSNR and SSIM and substantially lowers AMBE. These results indicate that the method is particularly effective for images with more complex local contrast deficiencies.

Overall, the results confirm that the adaptive and local nature of CLALHE offers an advantage over standard CLAHE in terms of structural preservation and balanced enhancement. The improvement is especially evident in PSNR and SSIM, while entropy behavior varies across images depending on how intensities are redistributed.

C. Discussion

The implementation results support the main idea behind CLALHE: image regions do not all require the same enhancement strength. By selecting parameters adaptively from histogram features and applying enhancement locally to subimages, the method better matches the enhancement process to local content.

One important observation is that better visual quality is not always associated with larger entropy. In some cases, CLAHE produces slightly higher entropy because it spreads intensity values more aggressively, but this may also result in noise amplification or over-enhancement. CLALHE, on the other hand, often yields a more visually natural output despite a moderate entropy value.

The main limitation of this implementation is that it was evaluated on only three input images. A larger and more diverse dataset would provide stronger statistical evidence. In addition, processing time was not analyzed extensively in this study, although the extra parameter-search stage may increase computational cost compared to baseline CLAHE.

VI. CONCLUSION

This paper presented a MATLAB implementation of Contrast Limited Adaptive Local Histogram Equalization for poor-contrast image enhancement. The method was implemented by combining adaptive histogram-based parameter estimation, CIQI-driven parameter selection, and local subimage enhancement. Experimental evaluation on three sample images showed that the implemented CLALHE approach generally outperforms conventional CLAHE in terms of PSNR, SSIM, and brightness preservation, while also producing visually more balanced results.

The study confirms that automatic parameter estimation can reduce dependence on manual tuning and improve enhancement consistency. The local processing strategy also helps preserve important details without excessively amplifying background regions. Future work can extend this implementation to color images, larger datasets, real-time optimization, and broader comparisons with other adaptive histogram equalization methods.

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